

Manipulating Data

1. Create EmpTest from Emp table by copying structure and data.

```
SQL> CREATE TABLE EmpTest11 AS SELECT * FROM Emp;
```

```
SQL> CREATE TABLE EmpTest11 AS  SELECT * FROM Emp;
Table created.
```

1. Add a new row into the EmpTest table for Empno, Ename, Sal columns. In Ename should have the Current User.

```
SQL> INSERT INTO EmpTest 11(Empno, Ename, Sal) VALUES (1234, 'Sowmya', 5000);
```

```
SQL> INSERT INTO EmpTest11 (Empno, Ename, Sal)
  2  VALUES (1234, 'Sowmya', 5000);
1 row created.
```

2. Update EmpTest by increasing the Salary of TURNER by 15%. Confirm your changes.

```
SQL> UPDATE EmpTest11
```

```
    SET Sal = Sal* 1.15
```

```
    WHERE Ename = 'TURNER';
```

```
SQL> UPDATE EmpTest11 SET Sal=Sal*1.5 where Ename='turner';
0 rows updated.
```

```
SQL> SELECT * FROM EmpTest11
```

```
    WHERE Ename='TURNER';
```

```
SQL> SELECT * FROM EmpTest11 WHERE Ename='TURNER';
```

EMPNO	ENAME	JOB	MGR	HIREDATE
103	TURNER	SALESMAN	105	08-SEP-81
1500	0	30		

3. Update the salary of Smith with salary of Scott using EmpTest table.

```
SQL> UPDATE EmpTest  SET Salary = (SELECT Salary FROM EmpTest
    WHERE Ename = 'SCOTT')  WHERE Ename = 'SMITH';
```

```
SQL> UPDATE EmpTest SET Sal=(SELECT Sal FROM EmpTest WHERE Ename='SCOTT') WH
ERE Ename='SMITH';

9 rows updated.
```

4. Increase all the employees salary by 10% in EmpTest table who are working in NEW YORK.

```
SQL> UPDATE EmpTest11 SET Sal= Sal* 1.10 WHERE Deptno IN
( SELECT Deptno FROM Dept WHERE Loc='NEW YORK' );
```

```
SQL> UPDATE EmpTest11
2 SET Sal = Sal * 1.10
3 WHERE Deptno IN (
4 SELECT Deptno
5 FROM Dept
6 WHERE Loc = 'NEW YORK'
7 );

4 rows updated.
```

5. Delete all the Comm data from EmpTest Table;

```
SQL>UPDATE EmpTest SET Comm = NULL;
```

```
SQL> UPDATE EmpTest11 SET Comm = NULL;

16 rows updated.
```

6. Delete all the employees from EmpTest table who are working in SALES dept.

```
SQL> DELETE FROM EmpTest WHERE Deptno =
( SELECT Deptno
FROM Dept
WHERE Dname='SALES'
);
```

```
SQL> DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM Dept WHERE Dname='SALES');

6 rows deleted.
```

7. Delete all the who are working with that employee, except that Employee.(Prompt for the ENAME).

```
SQL> DELETE FROM EmpTest
WHERE Deptno =
(SELECT Deptno
FROM EmpTest
```

WHERE Ename='&ENAME') AND Ename<>'&ENAME';

```
SQL> DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='&ENAME') AND Ename<>'&ENAME';
Enter value for ename: SMITH
Enter value for ename: SMITH
old 1: DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='&ENAME') AND Ename<>'&ENAME'
new 1: DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='SMITH') AND Ename<>'SMITH'
0 rows deleted.
```

8. Create Emp2 from Emp by only copying Empno,Ename,sal without copying data.

```
SQL> CREATE TABLE Emp2 AS
      SELECT Empno,Ename,Sal
      FROM Emp
      WHERE 1=2;
```

```
SQL>  CREATE TABLE Emp2 AS
      2      SELECT Empno,Ename,Sal
      3      FROM Emp
      4      WHERE 1=2;
Table created.
```

9. Create Emp3 from Emp by only copying Empno,Job without copying data.

```
SQL> CREATE TABLE Emp3 AS
      SELECT Empno,Job
      FROM Emp
      WHERE 1=2;
```

```
SQL> CREATE TABLE Emp3 AS
      2      SELECT Empno,Job
      3      FROM Emp
      4      WHERE 1=2;
Table created.
```

10. Using multitable insert Emp data into Emp2 and Emp3 Tables.

```
SQL> INSERT ALL
      INTO Emp2(Empno,Ename,Sal)
      VALUES(Empno,Ename,Sal)
      INTO Emp3(Empno,Job)
      VALUES(Empno,Job)
      SELECT Empno,Ename,Sal,Job
      FROM Emp;
```

```
SQL> INSERT ALL
  2
  3
  4
  5
  6
  7
      INTO Emp2(Empno,Ename,Sal)
      VALUES(Empno,Ename,Sal)
      INTO Emp3(Empno,Job)
      VALUES(Empno,Job)
      SELECT Empno,Ename,Sal,Job
      FROM Emp;

30 rows created.
```

11. Truncate Emp2 Table and insert following two rows as follows: 7788, SMITH, 4500 /7654, JACK, 3500.

SQL> TRUNCATE TABLE Emp2;

```
SQL> TRUNCATE TABLE Emp2;

Table truncated.
```

SQL>INSERT INTO Emp2 VALUES(7788,'SMITH',4500);

```
SQL> INSERT INTO Emp2 VALUES(7788,'SMITH',4500);

1 row created.
```

SQL>INSERT INTO Emp2 VALUES(7654,'JACK',3500);

```
SQL> INSERT INTO Emp2 VALUES(7654,'JACK',3500);

1 row created.
```

12. Commit the Data.

SQL>commit;

```
SQL> commit;

Commit complete.
```

13. Using Merge statement insert and update Emp2 using Emp.

SQL> MERGE INTO Emp2 E2

USING Emp E

ON (E2.Empno=E.Empno)

WHEN MATCHED THEN

UPDATE SET

E2.Ename=E.Ename,

E2.Sal=E.Sal

WHEN NOT MATCHED THEN

```
INSERT(Empno,Ename,Sal)
VALUES(E.Empno,E.Ename,E.Sal);
```

```
SQL>  MERGE INTO Emp2 E2
      2  USING Emp E
      3  ON (E2.Empno=E.Empno)
      4  WHEN MATCHED THEN
      5  UPDATE SET
      6  E2.Ename=E.Ename,
      7  E2.Sal=E.Sal
      8  WHEN NOT MATCHED THEN
      9  INSERT(Empno,Ename,Sal)
     10  VALUES(E.Empno,E.Ename,E.Sal);

15 rows merged.
```

14. Verify your changes.

```
SQL>select * from emp2;
```

```
SQL> select * from emp2;
```

EMPNO	ENAME	SAL
7788	SMITH	4500
7654	JACK	3500
113	ADAMSSMITH	1100
107	ALLEN	1600
108	WARDEN	1250
109	MARTINKING	1250
105	BLAKE	2850
110	JAMES	950
104	SMITHKO	800
115	RAMESH	2600
101	KING	5000
103	TURNER	1500
112	CLARKEN	2450
111	MILLERROBIN	1300
114	SARALA	2100
102	SCOTLER	3000
106	JONES	2975

```
17 rows selected.
```

15. Rollback the Data.

```
SQL>rollback;
```

```
SQL> rollback;

Rollback complete.
```

17.Using Merge statements update Emp2 table for only Empno=7788 and Insert only those employees whose salary is more than 3000.

```
SQL> MERGE INTO Emp2 E2
USING
(
SELECT *
FROM Emp
WHERE Sal>3000
) E
ON(E2.Empno=E.Empno)
WHEN MATCHED THEN
UPDATE SET
E2.Ename=E.Ename,
E2.Sal=E.Sal
WHERE E.Empno=7788
WHEN NOT MATCHED THEN
INSERT(Empno,Ename,Sal)
VALUES(E.Empno,E.Ename,E.Sal);
```

```
SQL> MERGE INTO Emp2 E2
2 USING
3 (
4 SELECT *
5 FROM Emp
6 WHERE Sal>3000
7 ) E
8 ON(E2.Empno=E.Empno)
9 WHEN MATCHED THEN
10 UPDATE SET
11 E2.Ename=E.Ename,
12 E2.Sal=E.Sal
13 WHERE E.Empno=7788
14 WHEN NOT MATCHED THEN
15 INSERT(Empno,Ename,Sal)
16 VALUES(E.Empno,E.Ename,E.Sal);
1 row merged.
```

18.Verify your changes.

```
SQL>select * from emp2;
```

```
SQL> select * from emp2;
```

EMPNO	ENAME	SAL
7788	SMITH	4500
7654	JACK	3500
101	KING	5000

19.Create a User WIPRO.

SQL> CREATE USER WIPRO IDENTIFIED BY wipro;

```
SQL> CREATE USER WIPRO IDENTIFIED BY wipro;
User created.
```

SQL> GRANT CONNECT, RESOURCE TO WIPRO;

```
SQL> GRANT CONNECT, RESOURCE TO WIPRO;
Grant succeeded.
```

20. Grant ALL permission on EMP table from SCOTT to WIPRO user.

SQL> GRANT ALL ON Emp TO WIPRO;

```
SQL> GRANT ALL ON EMP TO WIPRO;
Grant succeeded.
```

21. Delete all the employees in deptno 10 and do not issue a Commit.

Delete all the employees in deptno 10 and do not issue a Commit.

SQL> DELETE FROM Emp WHERE Deptno=10;

```
SQL> DELETE FROM Emp WHERE Deptno=10;
4 rows deleted.
```

22. From WIPRO user delete all the employees from SCOTT.EMP table in DEPTNO=10. What happens and Why?

SQL> DELETE FROM Scott.Emp WHERE Deptno=10;

```
SQL> DELETE FROM SCOTT.EMP WHERE DEPTNO = 10;
4 rows deleted.
```

23. Issue a Rollback in SCOTT user and Check the WIPRO user.

SQL> ROLLBACK;

```
SQL> ROLLBACK;
Rollback complete.
```

SQL> SELECT * FROM Scott.Emp WHERE Deptno=10;

```
SQL> SELECT * FROM Scott.Emp WHERE Deptno=10;

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    101  KING          PRESIDENT              17-NOV-81
    5000              10
    111  MILLERROBIN    CLERK                 101 23-JAN-82
    1300              10
    112  CLARKEN        MANAGER              101 09-JUN-81
    2450              10

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    114  SARALA          CLERK                 112 15-AUG-81
    2100              10
```

24. In SCOTT user give a query on EMP using FOR UPDATE clause with WAIT 20 seconds? What happens?

SQL> SELECT * FROM Emp FOR UPDATE WAIT 20;

```
SQL> SELECT * FROM Emp FOR UPDATE WAIT 20;

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    113  ADAMSSMITH    CLERK                 102 23-MAY-87
    1100              20
    101  KING          PRESIDENT              17-NOV-81
    5000              10
    102  SCOTLER      ANALYST              106 19-APR-81
    3000              20

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    103  TURNER        SALESMAN             105 08-SEP-81
    1500         0          30
    104  SMITHKO       CLERK                 106 17-DEC-80
    800          0          20
    105  BLAKE         MANAGER              101 01-MAY-81
    2850         0          30

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    106  JONES         MANAGER              101 02-APR-81
```

2975			20		
107	ALLEN		SALESMAN	105	20-FEB-81
1600	300	30			
108	WARDEN		SALESMAN	105	22-FEB-81
1250	500	30			
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
109	MARTINKING		SALESMAN	105	28-SEP-81
1250	1400	30			
110	JAMES		CLERK	105	03-DEC-81
950		30			
111	MILLERROBIN		CLERK	101	23-JAN-82
1300		10			
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
112	CLARKEN		MANAGER	101	09-JUN-81
2450		10			
114	SARALA		CLERK	112	15-AUG-81
2100		10			
115	RAMESH		ANALYST	112	10-OCT-81
2600		20			

15 rows selected.

25. In WIPRO User issue a ROLLBACK and now check in SCOTT user.

SQL>rollback;

```
SQL> rollback;
Rollback complete.
```

26. ROLLBACK all the transactions in SCOTT and WIPRO Users.

SQL>rollback;

```
SQL> rollback;
Rollback complete.
```